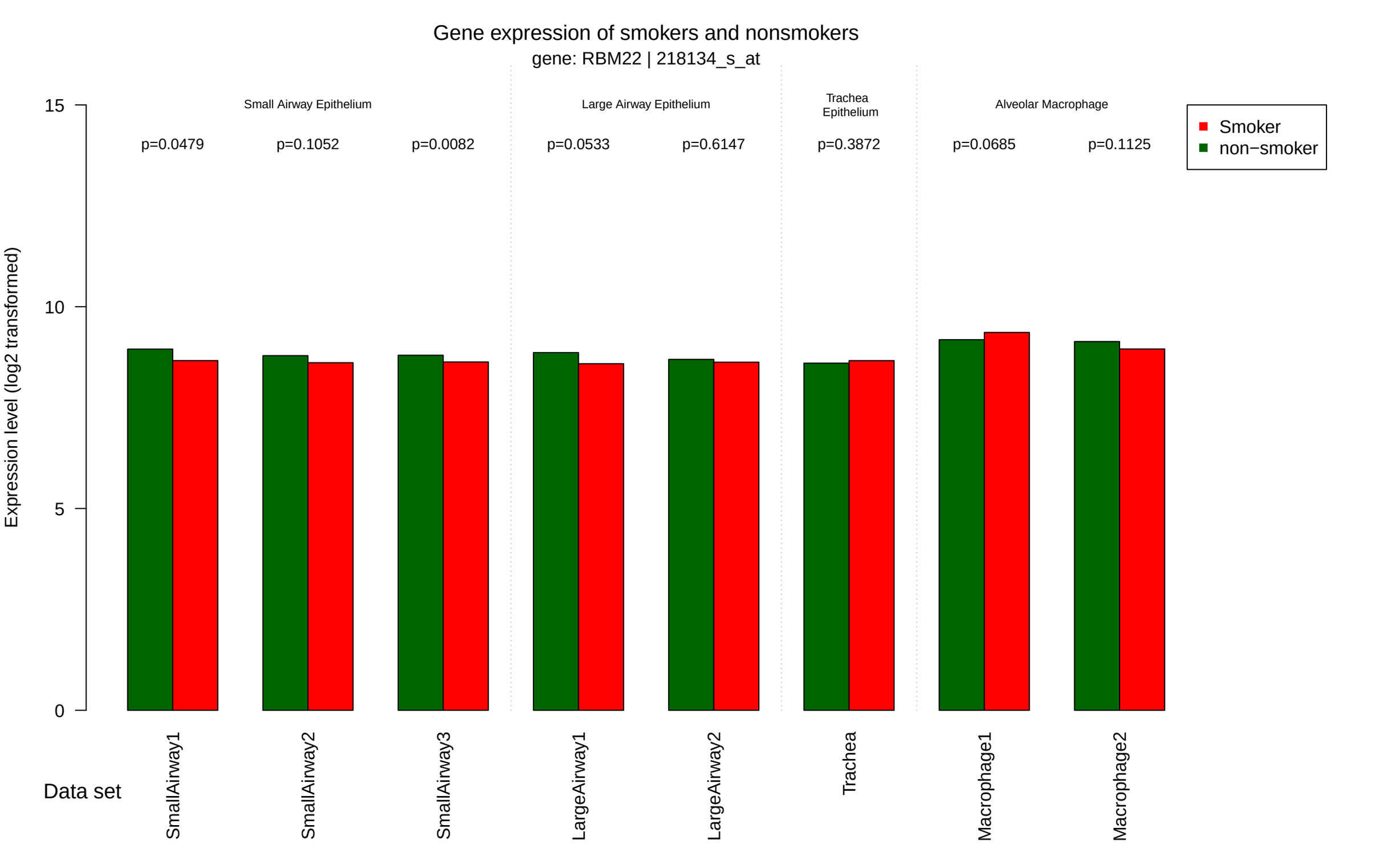
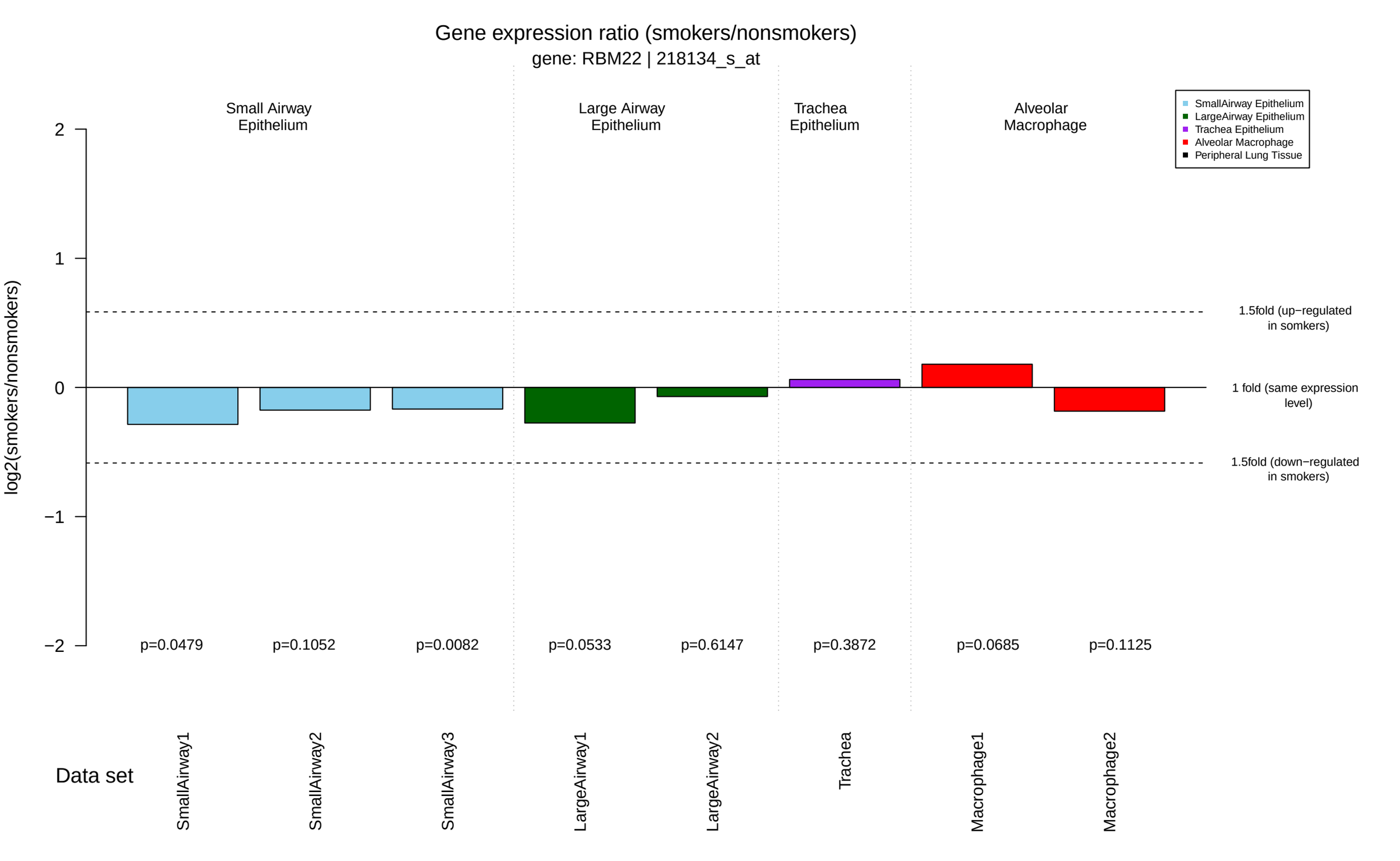
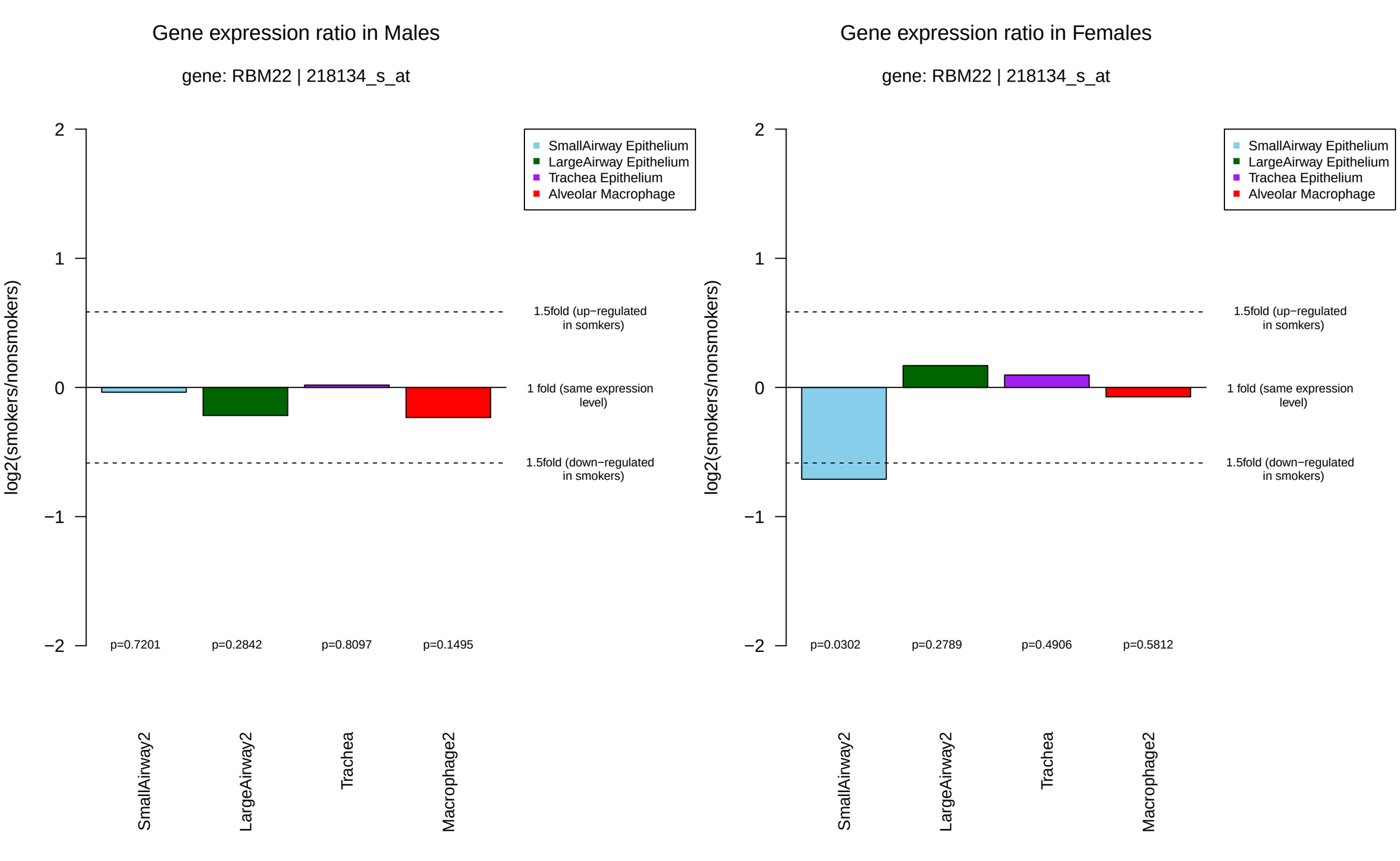
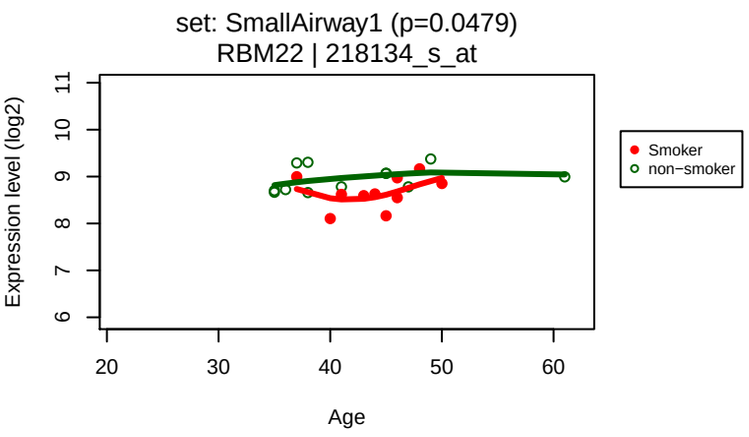
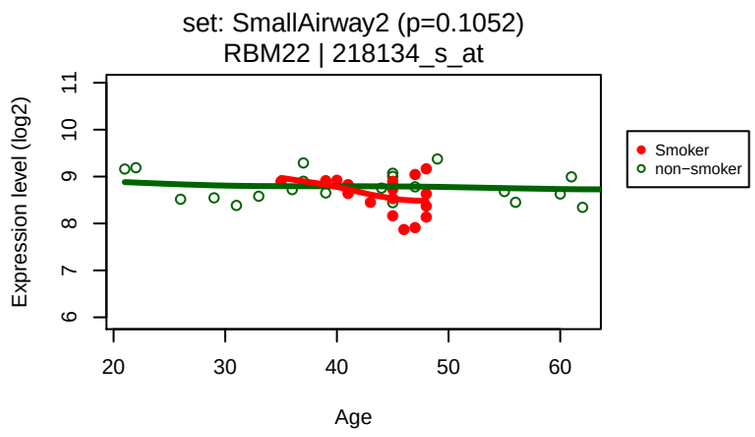
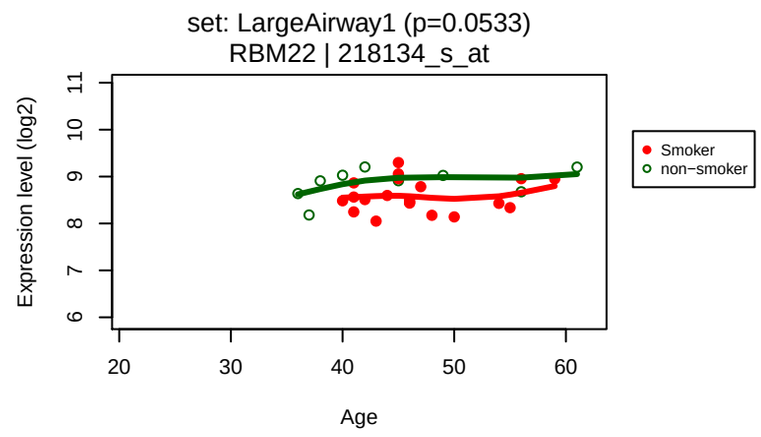

Gene expression of smokers and nonsmokers
gene: RBM22 | 218134_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0479
p=0.1052
p=0.0082
p=0.0533
p=0.6147
p=0.3872
p=0.0685
p=0.1125
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

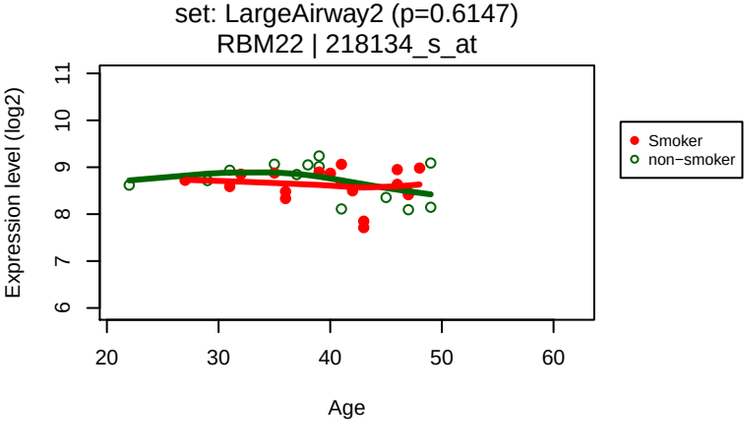

Gene expression ratio (smokers/nonsmokers)
gene: RBM22 | 218134_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0479
p=0.1052
p=0.0082
p=0.0533
p=0.6147
p=0.3872
p=0.0685
p=0.1125
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

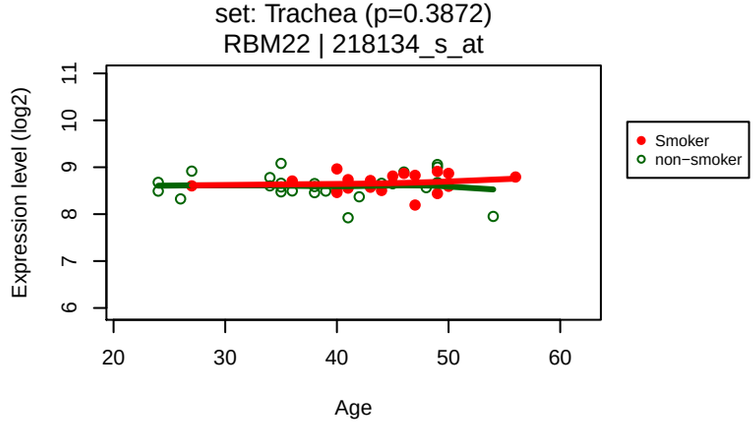

Gene expression ratio in Males
gene: RBM22 | 218134_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7201
p=0.2842
p=0.8097
p=0.1495
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBM22 | 218134_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0302
p=0.2789
p=0.4906
p=0.5812
SmallAirway2
LargeAirway2
Trachea
Macrophage2

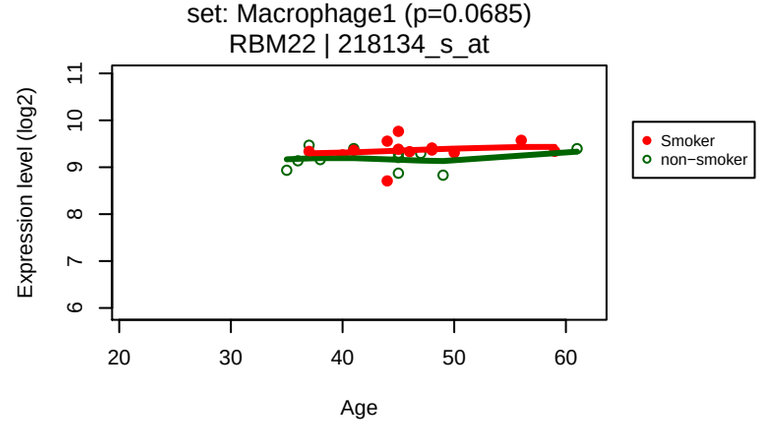

set: SmallAirway1 (p=0.0479)
RBM22 | 218134_s_at
Expression level (log2)
Age
Smoker
non−smoker

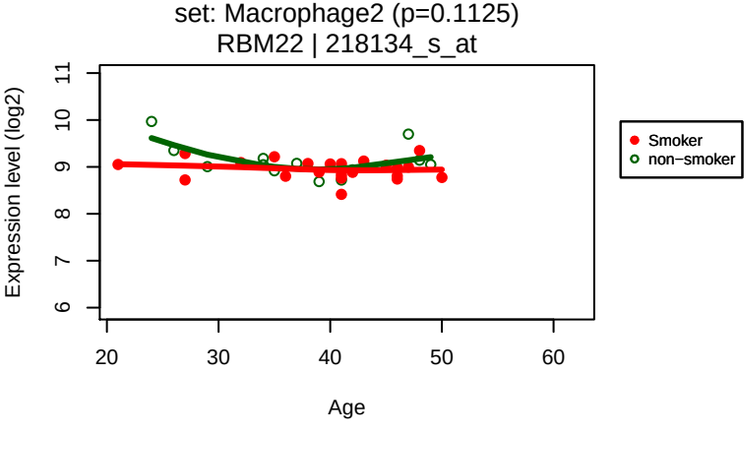

set: SmallAirway2 (p=0.1052)
RBM22 | 218134_s_at
Expression level (log2)
Age
Smoker
non−smoker

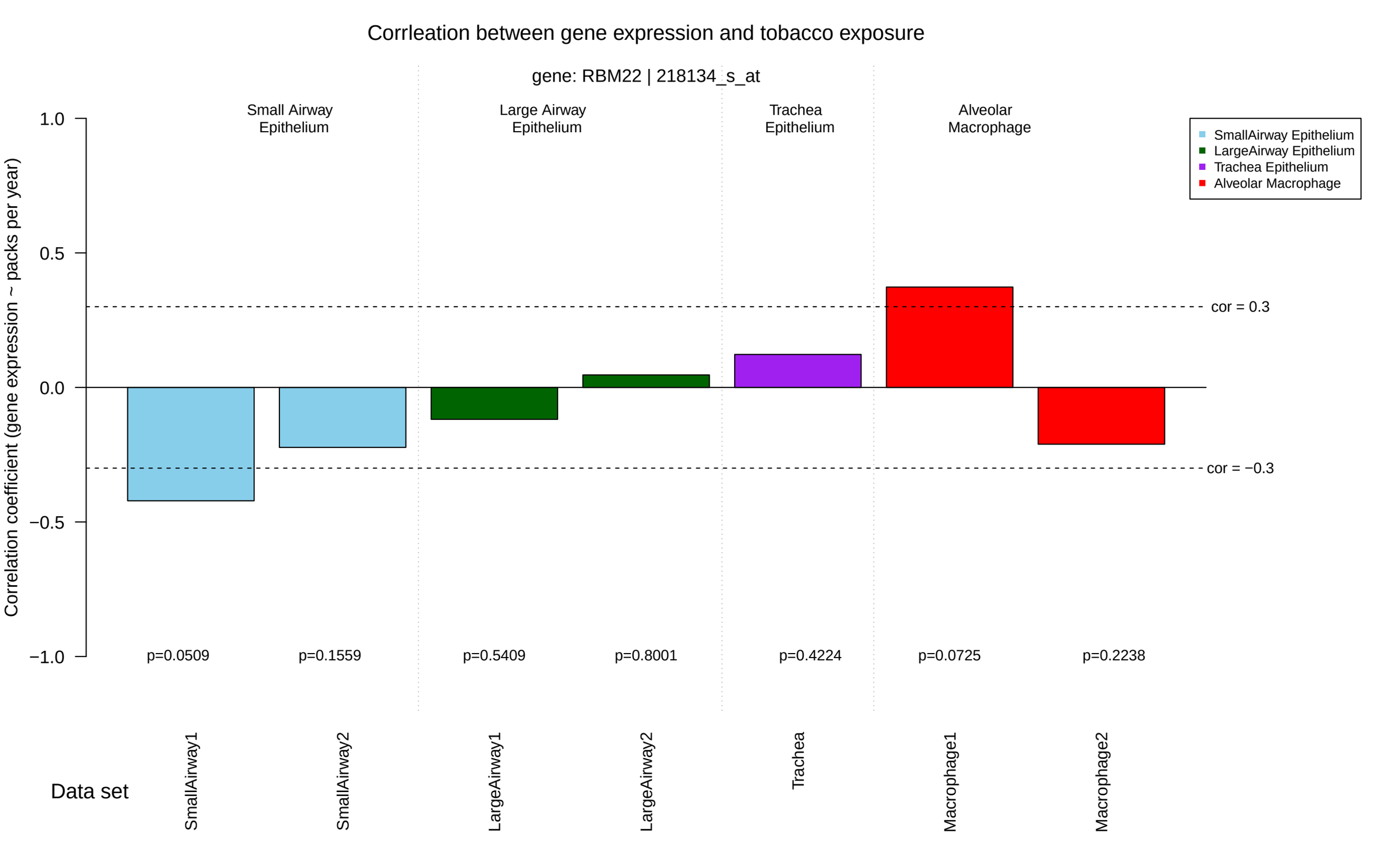

set: LargeAirway1 (p=0.0533)
RBM22 | 218134_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6147)
RBM22 | 218134_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3872)
RBM22 | 218134_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0685)
RBM22 | 218134_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1125)
RBM22 | 218134_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBM22 | 218134_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0509
p=0.1559
p=0.5409
p=0.8001
p=0.4224
p=0.0725
p=0.2238
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2